

DD_ASR-03 - Complaint - Developer Implementation Guide

Version: 1.0 Status: Draft for review Bundle: 10_other Audience: mid-level backend developers, ticketing developers, billing integration developers, QA Parent DD: DD_ASR-03 - Complaint

1. How To Read This Guide

Complaint is a TCK ticket flow. It is not a separate complaint engine.

The flow creates a TCK complaint ticket, loads ASR complaint rules, records an investigation, records resolution actions, optionally requests BIL adjustment, sends a customer-visible response, then resolves the TCK ticket.

ASR-03 can request actions in other modules. It does not directly change invoices, wallets, subscriptions, work orders, or customer master data.

2. Step 1 - Seed TCK Complaint Categories

Before any complaint can be created, seed TCK categories.

```
insert into ticket_category_config (
  category_id,
  operator_code,
  type_code,
  category_code,
  display_name,
  default_priority,
  default_queue_code,
  default_sla_policy,
  wo_allowed,
  default_wo_kind,
  status
) values
('tcat-billing-dispute', 'WIK', 'BILLING_COMPLAINT', 'BILLING_DISPUTE', 'Billing dispute', 'P2', 'BILLING_CARE', 'COMPLAINT', 'GENERAL', 'GENERAL_COMPLAINT', 'GENERAL COMPLAINT', 'P3', 'CARE_L1', 'COMPLAINT'),
('tcat-payment-dispute', 'WIK', 'BILLING_COMPLAINT', 'PAYMENT_DISPUTE', 'Payment dispute', 'P2', 'PAYMENT_OPS', 'COMPLAINT', 'GENERAL', 'GENERAL_COMPLAINT', 'GENERAL COMPLAINT', 'P3', 'CARE_L1', 'COMPLAINT'),
('tcat-repeat-tech', 'WIK', 'TECHNICAL_SUPPORT', 'REPEATED_TECHNICAL_FAULT', 'Repeated technical fault', 'P2', 'TECH_SUPPORT', 'COMPLAINT', 'GENERAL', 'GENERAL_COMPLAINT', 'GENERAL COMPLAINT', 'P3', 'CARE_L1', 'COMPLAINT'),
('tcat-install-delay', 'WIK', 'INSTALLATION_FOLLOWUP', 'INSTALLATION_DELAY_COMPLAINT', 'Installation delay complaint', 'P2', 'INSTALLATION', 'COMPLAINT', 'GENERAL', 'GENERAL_COMPLAINT', 'GENERAL COMPLAINT', 'P3', 'CARE_L1', 'COMPLAINT'),
('tcat-staff-conduct', 'WIK', 'GENERAL_INQUIRY', 'CONTRACTOR_OR_STAFF_CONDUCT', 'Contractor or staff conduct', 'P1', 'CAUSE', 'COMPLAINT', 'GENERAL', 'GENERAL_COMPLAINT', 'GENERAL COMPLAINT', 'P3', 'CARE_L1', 'COMPLAINT'),
('tcat-general-complaint', 'WIK', 'GENERAL_INQUIRY', 'GENERAL_COMPLAINT', 'General complaint', 'P3', 'CARE_L1', 'COMPLAINT', 'GENERAL', 'GENERAL_COMPLAINT', 'GENERAL COMPLAINT', 'P3', 'CARE_L1', 'COMPLAINT');
```

These are TCK categories. TCK owns ticket id, status, assignment, SLA, comments, and attachments.

3. Step 2 - Seed ASR Complaint Category Config

Now seed ASR behavior for those categories.

```
insert into asr_complaint_category_config (
  config_id,
  operator_code,
  category_code,
  default_type_code,
  default_priority,
  default_queue_code,
  investigation_required,
  supervisor_review_required,
  customer_response_required,
  compensation_allowed,
  source_modules_json,
  allowed_resolution_codes_json,
  status
) values
(
  'acc-billing-dispute',
  'WIK',
  'BILLING_DISPUTE',
  'BILLING_COMPLAINT',
  'P2',
  'BILLING_CARE',
  true,
```

```

    true,
    true,
    true,
    ['BIL-02-READ-01', "BIL-02-ADJ-01", "SIP-03", "DIS-0P-01"],
    ["VALID_COMPLAINT_CORRECTED", "PARTIALLY_VALID_CORRECTED", "NOT_VALID_EXPLAINED", "INCONCLUSIVE_EXPLAINED"],
    'ACTIVE'
  ),
  (
    'acc-repeat-tech',
    'WIK',
    'REPEATED_TECHNICAL_FAULT',
    'TECHNICAL_SUPPORT',
    'P2',
    'TECH_SUPPORT_L2',
    true,
    true,
    true,
    true,
    ['ASR-01', "W0-01", "PROV-INT-01"],
    ["VALID_COMPLAINT_CORRECTED", "PARTIALLY_VALID_CORRECTED", "NOT_VALID_EXPLAINED"],
    'ACTIVE'
  ),
  (
    'acc-staff-conduct',
    'WIK',
    'CONTRACTOR_OR_STAFF_CONDUCT',
    'GENERAL_INQUIRY',
    'P1',
    'CARE_SUPERVISOR',
    true,
    true,
    true,
    false,
    ['EM-01', "EM-02", "W0-01", "SALES-01", "TCK-01"],
    ["VALID_COMPLAINT_CORRECTED", "PARTIALLY_VALID_CORRECTED", "NOT_VALID_EXPLAINED", "ESCALATED_TO_MANAGEMENT"],
    'ACTIVE'
  );

```

Frontend and backend must read this config. Do not hardcode complaint rules per screen.

4. Step 3 - Customer Or Agent Creates Complaint Ticket

Backoffice calls TCK:

```

POST /api/tickets
Idempotency-Key: bo-complaint-20260604-000011
Content-Type: application/json

{
  "operatorCode": "WIK",
  "typeCode": "BILLING_COMPLAINT",
  "categoryCode": "BILLING_DISPUTE",
  "sourceChannel": "BACKOFFICE",
  "customerId": "CUS-100045",
  "accountId": "ACC-900045",
  "subscriptionId": "SUB-700045",
  "subject": "Customer disputes missing discount",
  "description": "Customer says the June invoice did not include a retention discount promised by care.",
  "linkedEntities": [
    {"entityType": "INVOICE", "entityId": "INV-2026-06-000045"},
    {"entityType": "SUBSCRIPTION", "entityId": "SUB-700045"}
  ],
  "customerVisible": true
}

```

TCK must:

1. validate typeCode and categoryCode;
2. create ticket;
3. create ticket_entity_link rows;
4. assign queue/SLA from ticket_category_config;
5. emit generic TicketCreated;
6. let ASR emit ComplaintCreated when the category maps to ASR-03.

Expected response:

```
{
  "ticketId": "tck-20260604-000011",
  "statusCode": "OPEN",
  "categoryCode": "BILLING_DISPUTE",
  "priorityCode": "P2",
  "assignedQueueCode": "BILLING_CARE"
}
```

5. Step 4 - Backoffice Loads Complaint Guidance

When the user opens the complaint, Backoffice calls:

```
GET /api/asr/complaints/tck-20260604-000011/guidance
```

Backend reads:

```
select *
from asr_complaint_category_config
where operator_code = 'WIK'
  and category_code = 'BILLING_DISPUTE'
  and status = 'ACTIVE';
```

Response:

```
{
  "ticketId": "tck-20260604-000011",
  "categoryCode": "BILLING_DISPUTE",
  "defaultQueueCode": "BILLING_CARE",
  "investigationRequired": true,
  "supervisorReviewRequired": true,
  "customerResponseRequired": true,
  "compensationAllowed": true,
  "sourceModules": ["BIL-02-READ-01", "BIL-02-ADJ-01", "SIP-03", "DIS-OP-01"],
  "allowedResolutionCodes": [
    "VALID_COMPLAINT_CORRECTED",
    "PARTIALLY_VALID_CORRECTED",
    "NOT_VALID_EXPLAINED",
    "INCONCLUSIVE_EXPLAINED"
  ],
  "allowedActions": [
    "START_INVESTIGATION",
    "ADD_FINDINGS",
    "CREATE_RESOLUTION_ACTION",
    "RECOMMEND_COMPENSATION",
    "ESCALATE",
    "SEND_CUSTOMER_RESPONSE",
    "RESOLVE"
  ]
}
```

The UI uses this response to decide visible buttons and required fields.

6. Step 5 - Start Investigation

The agent starts investigation:

```
POST /api/asr/complaints/tck-20260604-000011/investigation
Content-Type: application/json
```

```
{
  "customerClaim": "Customer says a 500 KES retention discount was promised and should appear on June invoice.",
  "assignedQueueCode": "BILLING_CARE",
  "sourceRefs": [
    { "module": "BIL-02-READ-01", "ref": "INV-2026-06-000045" },
    { "module": "SIP-03", "ref": "discount-assignment-8891" }
  ]
}
```

Insert:

```
insert into asr_complaint_investigation (
  investigation_id,
  ticket_id,
  operator_code,
  category_code,
  investigation_status,
  assigned_queue_code,
  assigned_user_id,
```

```

    source_refs_json,
    customer_claim,
    findings_summary,
    root_cause_code,
    validity_code,
    created_by_user_id,
    created_at
) values (
    'aci-001',
    'tck-20260604-000011',
    'WIK',
    'BILLING_DISPUTE',
    'OPEN',
    'BILLING_CARE',
    'usr-billing-014',
    ' [{"module": "BIL-02-READ-01", "ref": "INV-2026-06-000045"}, {"module": "SIP-03", "ref": "discount-assignment-8891"} ]',
    'Customer says a 500 KES retention discount was promised and should appear on June invoice.',
    null,
    null,
    null,
    'usr-care-001',
    now()
);

```

Then update TCK:

```

POST /api/tickets/tck-20260604-000011/assign
Content-Type: application/json

{
  "assignedQueueCode": "BILLING_CARE",
  "assignedUserId": "usr-billing-014",
  "statusCode": "ASSIGNED"
}

```

7. Step 6 - Gather Source Data

The complaint service calls owning module APIs. It must not read another module's tables directly.

For billing dispute:

```

GET /api/invoices/INV-2026-06-000045
GET /api/invoices/INV-2026-06-000045/lines
GET /api/discount-assignments/discount-assignment-8891
GET /api/discount-runtime/applications?invoiceId=INV-2026-06-000045

```

Example source summary stored in application memory or returned to UI:

```

{
  "invoice": {
    "invoiceId": "INV-2026-06-000045",
    "totalAmount": 4500.00,
    "currencyCode": "KES",
    "status": "ISSUED"
  },
  "discountAssignment": {
    "discountAssignmentId": "discount-assignment-8891",
    "status": "ACTIVE",
    "amount": 500.00,
    "validFrom": "2026-06-01",
    "validTo": "2026-06-30"
  },
  "runtimeApplication": {
    "found": false,
    "reason": "No discount runtime application record for invoice."
  }
}

```

If a source API fails:

1. keep investigation status WAITING_SOURCE;
2. update TCK status to WAITING_INTERNAL;
3. create a TCK internal comment naming the unavailable module;
4. retry through a worker or ask the owning team.

8. Step 7 - Complete Investigation

When facts are clear, the investigator completes:

```
POST /api/asr/complaints/investigations/aci-001/complete
Content-Type: application/json
```

```
{
  "findingsSummary": "The customer had an active discount assignment, but the discount runtime did not apply it to the Jun
  "rootCauseCode": "BILLING_DISCOUNT_RUNTIME_ERROR",
  "validityCode": "VALID",
  "sourceRefs": [
    {"module": "BIL-02-READ-01", "ref": "INV-2026-06-000045"},
    {"module": "DIS-0P-01", "ref": "discount-runtime-log-7712"},
    {"module": "SIP-03", "ref": "discount-assignment-8891"}
  ]
}
```

Update:

```
update asr_complaint_investigation
set investigation_status = 'COMPLETED',
  findings_summary = 'The customer had an active discount assignment, but the discount runtime did not apply it to the Jun
  root_cause_code = 'BILLING_DISCOUNT_RUNTIME_ERROR',
  validity_code = 'VALID',
  source_refs_json = '[{"module": "BIL-02-READ-01", "ref": "INV-2026-06-000045"}, {"module": "DIS-0P-01", "ref": "discount-runt
  completed_at = now()
where investigation_id = 'aci-001';
```

Emit:

```
{
  "eventType": "ComplaintInvestigationCompleted",
  "ticketId": "tck-20260604-000011",
  "investigationId": "aci-001",
  "rootCauseCode": "BILLING_DISCOUNT_RUNTIME_ERROR",
  "validityCode": "VALID"
}
```

9. Step 8 - Create Resolution Action

Because the complaint is valid, create an action.

```
POST /api/asr/complaints/tck-20260604-000011/resolution-actions
Content-Type: application/json
```

```
{
  "investigationId": "aci-001",
  "actionType": "BIL_ADJUSTMENT_REQUEST",
  "owningModule": "BIL-02-ADJ-01",
  "customerVisible": true,
  "actionSummary": "Request credit note for 500 KES missing discount."
}
```

Insert:

```
insert into asr_complaint_resolution_action (
  resolution_action_id,
  ticket_id,
  investigation_id,
  operator_code,
  action_type,
  action_status,
  owning_module,
  owning_ref_id,
  customer_visible,
  action_summary,
  created_by_user_id,
  created_at
) values (
  'acra-001',
  'tck-20260604-000011',
  'aci-001',
  'WIK',
  'BIL_ADJUSTMENT_REQUEST',
  'PLANNED',
  'BIL-02-ADJ-01',
  null,
  true,
  'Request credit note for 500 KES missing discount.',
  null,
  null
);
```

```

'usr-billing-014',
now()
);

```

10. Step 9 - Recommend Compensation

Create the complaint recommendation:

```

POST /api/asr/complaints/tck-20260604-000011/compensation-recommendations
Content-Type: application/json

```

```

{
  "reasonCode": "WRONG_DISCOUNT_APPLICATION",
  "recommendedDirection": "CREDIT",
  "recommendedScope": "LINE",
  "recommendedAmount": 500.00,
  "currencyCode": "KES",
  "sourceInvoiceId": "INV-2026-06-000045",
  "sourceLineRefs": [
    { "lineId": "INVLIN-001", "amountOverride": 500.00 }
  ],
  "justification": "Active approved discount was not applied to the issued invoice."
}

```

Insert:

```

insert into asr_complaint_compensation_recommendation (
  recommendation_id,
  ticket_id,
  operator_code,
  reason_code,
  recommended_direction,
  recommended_scope,
  recommended_amount,
  currency_code,
  source_invoice_id,
  source_line_refs_json,
  justification,
  status,
  adjustment_request_id,
  created_by_user_id,
  created_at
) values (
  'accr-001',
  'tck-20260604-000011',
  'WIK',
  'WRONG_DISCOUNT_APPLICATION',
  'CREDIT',
  'LINE',
  500.00,
  'KES',
  'INV-2026-06-000045',
  '["lineId":"INVLIN-001","amountOverride":500.00]',
  'Active approved discount was not applied to the issued invoice.',
  'DRAFT',
  null,
  'usr-billing-014',
  now()
);

```

Do not create a credit note here. ASR only records the recommendation.

11. Step 10 - Supervisor Review If Required

For BILLING_DISPUTE, config says supervisor review is required.

```

POST /api/asr/complaints/tck-20260604-000011/escalations
Content-Type: application/json

```

```

{
  "escalationLevel": "SUPERVISOR",
  "reasonCode": "COMPENSATION_REVIEW",
  "assignedQueueCode": "CARE_SUPERVISOR",
  "summary": "Review validated complaint and 500 KES credit recommendation."
}

```

Insert:

```

insert into asr_complaint_escalation (
  escalation_id,
  ticket_id,
  operator_code,
  escalation_level,
  reason_code,
  assigned_queue_code,
  assigned_user_id,
  status,
  decision_summary,
  created_by_user_id,
  created_at
) values (
  'ace-001',
  'tck-20260604-000011',
  'WIK',
  'SUPERVISOR',
  'COMPENSATION_REVIEW',
  'CARE_SUPERVISOR',
  null,
  'OPEN',
  null,
  'usr-billing-014',
  now()
);

```

Move ticket to review:

```

POST /api/tickets/tck-20260604-000011/status
Content-Type: application/json

{
  "statusCode": "UNDER_REVIEW",
  "reasonCode": "SUPERVISOR_REVIEW_REQUIRED"
}

```

When supervisor approves:

```

update asr_complaint_escalation
set status = 'APPROVED',
  assigned_user_id = 'usr-supervisor-002',
  decision_summary = 'Approved credit recommendation and customer response.',
  decided_at = now()
where escalation_id = 'ace-001';

```

12. Step 11 - Create BIL Adjustment Request

After approval, ASR calls BIL-02-ADJ-01:

```

POST /api/adjustments
Idempotency-Key: asr-complaint-tck-20260604-000011-credit
Content-Type: application/json

{
  "operatorCode": "WIK",
  "customerId": "CUS-100045",
  "accountId": "ACC-900045",
  "direction": "CREDIT",
  "scope": "LINE",
  "originalInvoiceId": "INV-2026-06-000045",
  "proposedLineItems": [
    {"parentLineItemId": "INVLINE-001", "amountOverride": 500.00}
  ],
  "reasonCode": "WRONG_DISCOUNT_APPLICATION",
  "externalCaseRef": "tck-20260604-000011",
  "description": "Credit requested from ASR-03 billing complaint."
}

```

Expected BIL response:

```

{
  "adjustmentRequestId": "adj-20260604-000007",
  "status": "PENDING_APPROVAL",
  "approvalWorkflowRef": "adjustment_approval_threshold_v1"
}

```

Patch ASR records:

```

update asr_complaint_compensation_recommendation
set status = 'REQUESTED',
  adjustment_request_id = 'adj-20260604-000007'

```

```

where recommendation_id = 'accr-001';

update asr_complaint_resolution_action
set action_status = 'REQUESTED',
    owning_ref_id = 'adj-20260604-000007'
where resolution_action_id = 'acra-001';

```

Important: BIL may still reject or request revision. ASR must show the BIL status but must not override BIL approval.

13. Step 12 - Send Customer Response

After investigation and required review, the agent sends the formal response:

```

POST /api/asr/complaints/tck-20260604-000011/customer-response
Content-Type: application/json

```

```

{
  "languageCode": "en",
  "responseChannel": "BACKOFFICE",
  "resolutionCode": "VALID_COMPLAINT_CORRECTED",
  "responseBody": "We reviewed your invoice and confirmed that your 500 KES discount was not applied correctly. A credit r
}

```

Insert:

```

insert into asr_complaint_customer_response (
  response_id,
  ticket_id,
  operator_code,
  language_code,
  response_body,
  response_channel,
  resolution_code,
  sent_by_user_id,
  sent_at
) values (
  'acr-001',
  'tck-20260604-000011',
  'WIK',
  'en',
  'We reviewed your invoice and confirmed that your 500 KES discount was not applied correctly. A credit request has been
  'BACKOFFICE',
  'VALID_COMPLAINT_CORRECTED',
  'usr-care-001',
  now()
);

```

Then add customer-visible TCK comment:

```

POST /api/tickets/tck-20260604-000011/comments
Content-Type: application/json

```

```

{
  "visibility": "CUSTOMER",
  "commentBody": "We reviewed your invoice and confirmed that your 500 KES discount was not applied correctly. A credit re
}

```

If the operator wants SMS/email, publish to NOT-01 with template COMPLAINT_RESPONSE_SENT.

14. Step 13 - Resolve The TCK Ticket

Before resolving, validate closure conditions.

```

-- Investigation exists and is complete.
select investigation_status
from asr_complaint_investigation
where ticket_id = 'tck-20260604-000011';

-- Customer response exists.
select response_id
from asr_complaint_customer_response
where ticket_id = 'tck-20260604-000011';

-- Required supervisor escalation is approved.
select status
from asr_complaint_escalation
where ticket_id = 'tck-20260604-000011'
    and reason_code = 'COMPENSATION_REVIEW';

```


Then call TCK:

```
POST /api/tickets/tck-20260604-000011/resolve
Content-Type: application/json

{
  "resolutionCode": "VALID_COMPLAINT_CORRECTED",
  "resolutionSummary": "Complaint validated. BIL adjustment request adj-20260604-000007 created for the missing discount."
}
```

TCK sets status to RESOLVED, stores resolution fields, emits TicketResolved, and ASR emits ComplaintResolved.

15. Alternative Flow - Complaint Not Valid

If investigation shows the invoice is correct:

1. Complete investigation with `validityCode = NOT_VALID`.
2. Create EXPLANATION resolution action.
3. Do not create compensation recommendation.
4. Send customer response explaining the finding.
5. Resolve with `NOT_VALID_EXPLAINED`.

Example final response:

```
{
  "resolutionCode": "NOT_VALID_EXPLAINED",
  "responseBody": "We reviewed the invoice and confirmed that the package charge and taxes match your active subscription"
}
```

16. Alternative Flow - Technical Complaint

If the complaint is repeated technical fault:

1. Create TCK complaint with category `REPEATED_TECHNICAL_FAULT`.
2. Load ASR guidance.
3. Start complaint investigation.
4. Pull previous ASR-01 diagnostics and WO support history.
5. If repair is needed, create/link ASR-01 or support WO using TCK/WO APIs.
6. Keep complaint ticket `WAITING_WORK_ORDER` if TCK policy says closure depends on WO result.
7. After WO result, complete investigation with root cause.
8. Recommend compensation only if policy and outage evidence support it.
9. Send formal customer response.
10. Resolve complaint.

The complaint ticket tracks dissatisfaction and formal response. ASR-01/WO track technical restoration.

17. Suggested Worker Commands

These run inside TCK/ASR workers and can be consolidated into one worker process with multiple scheduled commands.

```
asr-complaint:sla-scan
asr-complaint:duplicate-scan
asr-complaint:source-api-retry
asr-complaint:bil-adjustment-status-sync
asr-complaint:escalation-reminder
```

Command	Purpose
<code>asr-complaint:sla-scan</code>	Finds complaints approaching/breaching SLA and emits alerts.
<code>asr-complaint:duplicate-scan</code>	Links repeated complaints by same customer/source/category.

Command	Purpose
asr-complaint:source-api-retry	Retries source API lookups that left investigation in WAITING_SOURCE.
asr-complaint:bil-adjustment-status-sync	Reads BIL adjustment status for linked compensation recommendations.
asr-complaint:escalation-reminder	Reminds supervisor queues about open escalation records.

18. Error Codes

Code	Meaning
ASR_COMP_CATEGORY_NOT_CONFIGURED	TCK category has no ASR complaint config.
ASR_COMP_INVESTIGATION_ALREADY_EXISTS	Ticket already has an investigation.
ASR_COMP_INVESTIGATION_REQUIRED	Cannot resolve because investigation is required and missing.
ASR_COMP_INVESTIGATION_NOT_COMPLETE	Cannot resolve because investigation is still open.
ASR_COMP_CUSTOMER_RESPONSE_REQUIRED	Cannot resolve without customer-visible response.
ASR_COMP_SUPERVISOR_REVIEW_REQUIRED	Cannot proceed until supervisor review is approved.
ASR_COMP_COMPENSATION_NOT_ALLOWED	Category does not allow compensation recommendation.
ASR_COMP_BIL_ADJUSTMENT_FAILED	BIL adjustment request failed.
ASR_COMP_INVALID_RESOLUTION_CODE	Resolution code is not allowed for category.
ASR_COMP_INTERNAL_TEXT_IN_CUSTOMER_RESPONSE	Response contains blocked internal-only wording.

19. Minimum Implementation Order

1. Seed TCK complaint categories.
2. Create ASR complaint tables and indexes.
3. Seed ASR complaint category config.
4. Implement guidance API.
5. Implement investigation start/complete APIs.
6. Implement source API adapter pattern for BIL, PAY-GW, WO, FUL, OSR, EM.
7. Implement resolution-action API.
8. Implement compensation recommendation API.
9. Implement BIL adjustment request integration.
10. Implement escalation API.
11. Implement customer-response API.
12. Patch TCK resolve validation to call ASR complaint closure checks.
13. Emit ASR complaint events.
14. Patch Backoffice complaint screens.
15. Patch customer-safe self-care status/response view if enabled.
16. Add E2E tests.

20. Test Scenarios

20.1 Valid Billing Complaint With Credit Request

Input:

- Complaint category BILLING_DISPUTE.
- Invoice linked.

- Investigation finds missing discount.

Expected:

- Investigation completed as VALID.
- Compensation recommendation created.
- Supervisor review approved.
- BIL adjustment request created.
- Customer response sent.
- TCK ticket resolved.

20.2 Billing Complaint Not Valid

Input:

- Customer disputes invoice.
- Investigation finds invoice matches package/tax configuration.

Expected:

- Investigation completed as NOT_VALID.
- No BIL adjustment request.
- Customer response explains finding.
- Ticket resolves as NOT_VALID_EXPLAINED.

20.3 Staff Conduct Complaint

Input:

- Category CONTRACTOR_OR_STAFF_CONDUCT.

Expected:

- Supervisor review required.
- Compensation disabled unless separately configured.
- Internal findings hidden from customer.
- Customer receives professional response.

20.4 Repeated Technical Fault Complaint

Input:

- Category REPEATED_TECHNICAL_FAULT.
- Previous ASR-01 tickets and WO records exist.

Expected:

- Investigation reads ASR-01/WO history.
- Support WO or ASR-01 link created if needed.
- Complaint waits for WO result if policy requires.
- Customer response references outcome, not internal diagnostics.

20.5 Closure Blocked

Input:

- User attempts to resolve complaint before investigation or customer response.

Expected:

- API returns 409 ASR_COMP_INVESTIGATION_REQUIRED or 409 ASR_COMP_CUSTOMER_RESPONSE_REQUIRED.
- TCK status remains open.

21. Developer Checklist

- Complaint flow uses TCK ticket lifecycle.
- ASR complaint tables do not duplicate TCK status, assignment, or comments.
- Source data is pulled through owning module APIs only.
- Investigation records root cause and validity.
- Customer response is stored and also added as TCK customer-visible comment.
- Financial correction is requested through BIL-02-ADJ-01 only.
- Closure validation blocks incomplete complaints.
- Events are emitted for reporting and timeline.
- Worker commands can run inside the existing TCK worker pool.